SIMATS ENGINEERING
ASSESSMENT TOOL 1
COURSE NAME: Artificial Intelligence COURSE CODE: CSA1741

CO1-AT1-Analytical Problem Solving

Register Number	**Bandla Brahmendra babu**
Name	**192312439**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks: 50**

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

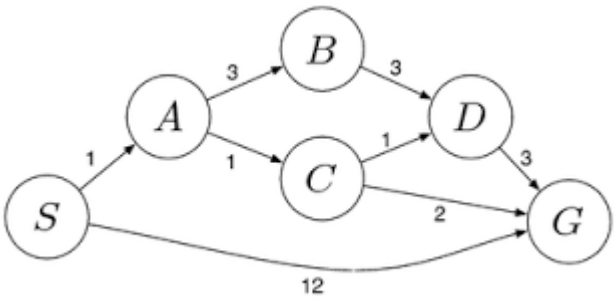
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path

from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

*I **Brahmendra babu(192312439)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Brahmendra

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

Question 1

Let (x, y) represent the amount of water in the 4-gallon jug and the 3-gallon jug, respectively. The initial state is (0, 0). The capacity limits are $0 \leq x \leq 4$ and $0 \leq y \leq 3$. The permitted actions are filling a jug from the pump, emptying a jug onto the ground, and pouring water between the jugs until the source is empty or the destination is full.
The goal is to reach a state where the 4-gallon jug contains exactly 2 gallons. This can be achieved by using the 3-gallon jug to create a remainder of 2 gallons.

Step	**Action performed**	**State (4-gallon, 3-gallon)**	**Explanation**
1	Fill the 3-gallon jug	(0, 3)	The small jug contains 3 gallons.
2	Pour it into the 4-gallon jug	(3, 0)	All 3 gallons are transferred.
3	Fill the 3-gallon jug again	(3, 3)	Both jugs contain 3 gallons.
4	Pour into the 4-gallon jug until full	(4, 2)	Only 1 gallon fits in the larger jug, leaving 2 gallons in the smaller jug.
5	Empty the 4-gallon jug	(0, 2)	The smaller jug still holds 2 gallons.
6	Transfer the remaining water to the 4-gallon jug	(2, 0)	The required quantity is obtained.

Explanation and result
The key step is the fourth one. Since the 4-gallon jug already contains 3 gallons, it can accept only 1 additional gallon. Pouring from the full 3-gallon jug therefore leaves exactly 2 gallons in the smaller jug. After emptying the larger jug, the remaining 2 gallons can be transferred into it.
Final state: (2, 0). The 4-gallon jug contains exactly 2 gallons. This is a state-space search problem: jug configurations are states, filling and pouring operations are actions, and the goal test checks the quantity in the larger jug. The method follows all rules and uses no additional measuring device.

ANSWER 2: MARS ROVER AS AN INTELLIGENT AGENT

A Mars Rover is a robotic system that explores the Martian surface, observes terrain, investigates rocks and soil, and sends scientific information to Earth. It is an intelligent agent because it senses its environment and selects actions to achieve mission goals.

Percepts

Percepts are inputs received from the environment through sensors. The rover uses:
Camera images showing rocks, slopes, paths and hazards.
Distance and terrain information to estimate obstacle locations and safe routes.
Scientific measurements about the properties and composition of rocks and soil.
Environmental readings such as temperature, pressure, lighting and dust conditions.
Internal-state information such as estimated position, orientation, wheel movement, battery level and instrument status.
Communication information indicating whether data can be transmitted to Earth.

Operating environment

Partially observable: sensors provide limited views and hidden hazards may remain undetected.
Stochastic in practice: movement may be uncertain because of loose soil, wheel slippage or uneven terrain.
Sequential: each movement changes the rover's position and affects later decisions.
Dynamic: lighting, temperature, dust and other conditions may change.
Continuous: position, orientation, speed and sensor readings vary continuously.
Primarily single-agent: the rover plans its own actions, although mission control may provide instructions.
Resource-constrained: energy, time, communication opportunities and instrument availability limit its activities.

Actions and performance

Move forward or backward and turn toward a target.
Stop or choose an alternate route when an obstacle or unsafe slope is detected.
Capture images and operate scientific instruments.
Inspect or collect samples using available equipment.
Store and transmit useful data to Earth.
Conserve energy and enter safe mode when required.
Re-plan the route when new observations show that the current plan is unsafe.
Performance can be measured by the quality of scientific data, targets reached, obstacles avoided, equipment reliability, energy efficiency, time efficiency, successful communication and completion of mission objectives.

Suitable agent architecture

A model-based, goal-based agent with utility-based decision-making is appropriate. The model-based component maintains an internal representation of location, terrain, obstacles and equipment condition. The goal-based component plans actions to reach scientific targets. The utility-based component compares alternatives according to scientific value, safety, time, energy and risk.
For example, if the shortest route to a rock crosses a steep slope, the rover may choose a longer but safer route. Combining perception, internal-state modelling, goal-directed planning and utility evaluation helps the rover explore uncertain terrain safely and efficiently.

ANSWER 3: EIGHT QUEENS PROBLEM

The objective is to place eight queens on an 8 × 8 chessboard so that no queen attacks another. A queen attacks along its row, column and diagonals. Therefore, every pair of queens must occupy different rows and columns and must not share a diagonal.

Problem formulation

Initial state: an empty chessboard.
State representation: a partial arrangement of queens; a common method records the row selected for each column.
State space: all possible placements considered by the search process.
Actions: place a queen in a square that does not conflict with existing queens.
Transition model: add a queen to the chosen square.
Goal test: eight queens are placed with no shared row, column or diagonal.
Path cost: if each placement costs one unit, a complete solution uses eight placements; the objective is to find a valid arrangement.

Backtracking search

Backtracking is a depth-first search technique that builds a solution incrementally. It places one queen in each column, starting from the first column. After each placement, it checks whether the queen conflicts with any previously placed queen. If safe, it proceeds to the next column. If no row is safe, it returns to the previous column and changes that queen's position.
Begin with an empty board.
Select a row in the first column.
Check for row and diagonal conflicts.
If safe, move to the next column.
If unsafe, try another row.
If no row works, backtrack to the previous column.
Continue until all eight queens are placed safely.

One valid solution

The vector uses the index as the column and the value as the row:
[1, 5, 8, 6, 3, 7, 2, 4]
The queens are at (column, row) positions (1,1), (2,5), (3,8), (4,6), (5,3), (6,7), (7,2) and (8,4).

Verification and result

Each column contains one queen and all row values are different, so no two queens share a row or column. Two queens share a diagonal only when the absolute difference between their row numbers equals the absolute difference between their column numbers. The listed arrangement avoids these conflicts and is therefore valid.
Backtracking reduces unnecessary work by rejecting a partial arrangement as soon as a conflict appears, instead of waiting until all eight queens are placed. The vector [1, 5, 8, 6, 3, 7, 2, 4] is one valid solution.

ANSWER 4: OLA CAB BOOKING – INTELLIGENT AGENT

A customer using a cab-booking application enters a pickup location, destination and preferred cab category. The application identifies suitable available cabs, estimates fare and arrival time, presents options and completes the booking only after customer confirmation.

Agent type and problem formulation

A goal-based problem-solving agent with cost-based evaluation is suitable. Its goal is to arrange a ride from pickup to destination while considering cab availability and customer preferences.
Initial state: the customer has opened the application but has not confirmed a ride.
State information: pickup, destination, preferred category, available drivers, estimated fare, arrival time and route details.
Actions: enter locations, select a category, search for cabs, compare options, select a cab, confirm or cancel.
Goal state: a suitable cab is booked and driver details are provided.
Constraints: locations must be valid, the cab must be available, and the customer must approve the booking details.

Pseudocode

```
BEGIN
INPUT pickup_location
INPUT destination
DISPLAY available cab categories
INPUT preferred_cab_type
FIND available cabs matching preferred_cab_type
IF no matching cab is available THEN
DISPLAY unavailability message
OFFER other available cab categories
ELSE
FOR each suitable available cab DO
ESTIMATE fare and arrival time
CHECK route feasibility
END FOR
DISPLAY cab options and estimates
INPUT customer_selection
DISPLAY selected cab and ride details
ASK customer for confirmation
IF customer confirms THEN
CREATE booking
DISPLAY driver and arrival details
ELSE
CANCEL booking request
END IF
END IF
END
```

Working and performance

Collect pickup and destination details.
Ask the customer to select a preferred cab category.
Search for available cabs matching that category.
Estimate fare and arrival time for each suitable option.
Display available options and their details.
Let the customer select an option and review the details.
Create the booking and show driver details only after confirmation; otherwise cancel the request.
Performance can be evaluated using booking success rate, accuracy of fare and arrival estimates, waiting time, suitability of the selected cab, customer satisfaction and failed bookings. The agent should respect the customer's preferences and must not finalize a booking without confirmation.
Conclusion: The goal-based agent treats cab booking as a problem with an initial state, actions, constraints and a goal state. Cost and time estimates help compare choices, and customer confirmation completes the process.

ANSWER 5: UNIFORM COST SEARCH (UCS)

Uniform Cost Search is an uninformed search method that expands the frontier path with the smallest cumulative cost g(n). It is suitable for a weighted graph because routes may have different costs. A priority queue keeps candidate paths ordered by their total accumulated costs.

Graph edges and costs
S → A = 1
S → G = 12
A → B = 3
A → C = 1
B → D = 3
C → D = 1
C → G = 2
D → G = 3

Search process
The search begins at S with cost zero. Expanding S adds S → A at cost 1 and S → G at cost 12. Since S → A has the lower cost, UCS expands it first. From A, it can reach C at total cost 2 and B at total cost 4. UCS continues to expand the least-cost frontier paths.

Stage	**Path selected or considered**	**Cost**	**Explanation**
1	S	0	Initial node.
2	S → A	1	A is reached from S at cost 1.
3	S → A → C	2	C is reached through A at total cost 2.
4	S → A → C → D	3	D is reached through C at total cost 3.
5	S → A → C → G	4	G is reached through C at total cost 4.

Comparison of candidate routes

Path	**Calculation**	**Total cost**
S → G	12	12
S → A → C → G	1 + 1 + 2	4
S → A → C → D → G	1 + 1 + 1 + 3	6
S → A → B → D → G	1 + 3 + 3 + 3	10

Optimal solution and interpretation
The least-cost path is S → A → C → G.

Minimum total cost = 1 + 1 + 2 = 4 units

The direct route S → G costs 12 units. The route through A, C and D costs 6 units, while the route through A, B and D costs 10 units. The route S → A → C → G has the lowest cost among these routes, at 4 units.

UCS compares accumulated path cost rather than simply counting edges. Assuming non-negative edge costs, it finds an optimal path when the goal is selected for expansion from the priority queue.

Conclusion: The optimal delivery route is S → A → C → G, with a minimum total travel cost of 4 units.